Shape Classification from Noisy 2D Perimeter Point Clouds via Multi-Hypothesis Fitting and Derivative-Based Straightness

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Abstract

We address the problem of classifying a noisy 2D point cloud into one of four shape families—circle, ellipse, square, or rectangle—given *only* the (x, y) coordinate pairs of the sampled points as input. Rather than training a classifier from labeled examples, we implement a fully deterministic multi-hypothesis pipeline that fits all four candidate models to every input and selects the class with the smallest adjusted residual score. Circles are fit using the Kasa algebraic least-squares method. Ellipses are fit using the Fitzgibbon–Pilu–Fisher direct least-squares method, which solves a generalized eigenvalue problem subject to an ellipticity constraint. Squares are fit by first applying a circle fit to the centered point cloud and then dividing the resulting radius by an empirically calibrated constant—determined by fitting circles to thousands of known squares—to recover the side length, followed by a brute-force rotation search. Rectangles are fit by Nelder–Mead optimization over width, height, and angle, initialized via PCA. A derivative-based sectional straightness score and a curvature-consistency corner score are incorporated as proportional bonuses to separate curved from rectilinear shapes under noise. We evaluate on a synthetic benchmark of 3,600 shapes spanning 36 noise levels from 0.00 to 0.70 (as a fraction of the shape's characteristic dimension). The deterministic method achieves 100% family-matching accuracy up to noise level 0.20 and outperforms a Random Forest baseline at all noise levels below approximately 0.32. Across all noise levels, the deterministic method achieves 84.8% family-matching accuracy versus 92.6% for the Random Forest; however, noise levels above approximately 0.30 produce shapes that are largely indistinguishable even to a human observer, making the deterministic method's performance in the practical regime strong.

1 Introduction

Classifying geometric shapes from noisy point data arises frequently in computer vision, robotics, and scientific measurement, where observations of object boundaries are sparse and corrupted. While deep learning dominates many classification tasks, problems with a small, geometrically well-understood hypothesis class offer an alternative: fit every candidate model explicitly and let the residuals decide. This paper develops such an approach for four shape families—circle, ellipse, square, and rectangle—applied to 2D perimeter point clouds.

Problem statement. Given an unordered set of (x, y) coordinate pairs $P = \{p_i\}_{i=1}^{N} \subset \mathbb{R}^2$ sampled from the boundary of an unknown planar shape belonging to one of $\{\text{circle}, \text{ellipse}, \text{square}, \text{rectangle}\}$ and perturbed by additive noise, predict the underlying class. Crucially, the coordinate pairs are the *only* input; no labels, metadata, or prior information about the shape is provided to the classifier.

Approach. For every input point cloud we simultaneously fit all four candidate models, compute an adjusted residual score for each, and output the class with the lowest score. No candidate is ever skipped or pre-filtered; all four always compete. To improve discrimination between curved

and rectilinear shapes at moderate-to-high noise, we compute a derivative-based straightness score from the local finite-difference slopes of consecutive points and incorporate it as a proportional bonus in the adjusted residual.

Why family-matching accuracy. Our classifier always outputs one of the four specific labels. However, there is an inherent geometric ambiguity within each family. An ellipse fit applied to data that was generated from a circle will simply find two nearly equal semi-axes—it fits perfectly well because a circle is a special case of an ellipse. Similarly, a rectangle fit applied to square data will find nearly equal width and height, since a square is a special case of a rectangle. The more general model naturally subsumes the special case, and no information is lost. This means that within the curved family (circle and ellipse) and within the rectilinear family (square and rectangle), the within-family distinction can become geometrically ambiguous under noise. We therefore report *family-matching accuracy*—which counts a prediction as correct whenever the predicted class belongs to the same geometric family as the ground truth. Family matching measures whether the method correctly identifies the *boundary type* (curved vs. straight-edged). It is used *only* for evaluation; the classifier itself always fits and scores all four models independently.

Noise levels and practical relevance. In our benchmark, noise is expressed as a *fraction of the shape's characteristic dimension* (radius for circles/ellipses, side length for squares/rectangles). A noise level of 0.20 means that each point is perturbed by a random displacement of up to 20% of the shape's size. At noise levels above approximately 0.30, shapes become visually indistinguishable—a square with 30% of its side length added as random jitter to each point looks essentially like a noisy blob. In virtually all practical applications (manufactured parts, satellite imagery, medical imaging), noise levels are well below this threshold, making the deterministic method's performance in the low-to-moderate regime highly relevant.

Contributions.

- A training-free, deterministic classifier for four shape families, operating on raw (x, y) coordinates alone.
- Circle fitting via the Kasa algebraic method with a proof that the normal equations minimize the algebraic residual.
- Ellipse fitting via the Fitzgibbon–Pilu–Fisher method with a proof of the conic discriminant condition.
- Square fitting via an empirically calibrated circle-to-square radius conversion and brute-force rotation search.
- Rectangle fitting via PCA-initialized Nelder–Mead optimization, with a full derivation of how PCA determines the initial angle.
- A scale-invariant, derivative-based straightness score incorporated as a proportional bonus.
- An empirical comparison against a Random Forest baseline, demonstrating that the deterministic method *outperforms* the Random Forest at all noise levels below ≈ 0.32 and achieves perfect accuracy up to noise level 0.20.

2 Related Work

Shape classification from point data draws on two traditions: shape descriptors fed to classifiers, and direct parametric model fitting. Classical descriptors include Hu moments [Hu, 1962], circularity, and compactness. Modern approaches employ point-cloud neural networks such as PointNet [Qi et al., 2017]. On the model-fitting side, algebraic circle fitting is well established [Kasa, 1976, Pratt, 1987, Taubin, 1991]. For ellipses, the Fitzgibbon–Pilu–Fisher direct least-squares method [Fitzgibbon et al., 1999] guarantees an elliptic solution via a constrained eigenvalue problem. Line and polygon fitting via RANSAC [Fischler and Bolles, 1981] and Hough transforms provide outlier robustness. PCA-based orientation estimation is standard for initializing oriented bounding boxes [O'Rourke, 1998].

Our approach applies parametric fits for all four classes simultaneously and supplements the residuals with a derivative-based straightness diagnostic. Unlike learned methods, it requires no training data and produces interpretable fitted parameters alongside the class label.

3 Problem Setup and Notation

Let $P = \{p_i\}_{i=1}^N$ with $p_i = (x_i, y_i)^\top \in \mathbb{R}^2$. The input to the classifier is *only* these coordinate pairs. Each point is assumed to have been sampled from the true perimeter $\partial S^\star$ of one of the four shape families and then corrupted by additive uniform noise. Concretely, a noise level of η means each coordinate is displaced by a uniform random value in $[-\eta \cdot L, \eta \cdot L]$, where L is the shape's characteristic size (radius for circles, semi-axis for ellipses, side length for squares, or the relevant side for rectangles).

For a candidate shape family $\mathcal{F}$ with fitted parameters $\hat{\theta}$, the point-to-boundary distance is $d(p, \partial S(\hat{\theta})) := \inf_{q \in \partial S(\hat{\theta})} \|p - q\|_2$, and the raw residual is $R(\mathcal{F}; P) := \sum_{i=1}^N d(p_i, \partial S(\hat{\theta}))$.

4 Method

4.1 Multi-Hypothesis Fitting and Selection

Given input P (a set of (x, y) pairs), we compute fits for all four families and output the class with the lowest adjusted score:

$$\widehat{\mathcal{F}}(P) = \arg \min_{\mathcal{F} \in \{\circ, \mathrm{e}, \square, \mathrm{r}\}} \widetilde{R}(\mathcal{F}; P).$$

No candidate is skipped; all four always compete.

4.2 Circle Fitting by Algebraic Least Squares

We use the Kasa algebraic least-squares method [Kasa, 1976]. From $(x - a)^2 + (y - b)^2 = r^2$, expanding and rearranging with $F = a^2 + b^2 - r^2$ gives the linear system $A\beta = s$ where

$$A = \begin{bmatrix} -2x_1 & -2y_1 & 1 \\ \vdots & \vdots & \vdots \\ -2x_N & -2y_N & 1 \end{bmatrix}, \quad \beta = \begin{bmatrix} a \\ b \\ F \end{bmatrix}, \quad s = \begin{bmatrix} -(x_1^2 + y_1^2) \\ \vdots \\ -(x_N^2 + y_N^2) \end{bmatrix}.$$

Solving $A^\top A\,\beta = A^\top s$ yields the center (a, b) and F; the radius is $r = \sqrt{a^2 + b^2 - F}$.

Proposition 1 (Normal equations minimize the algebraic residual)**.** *Let $A \in \mathbb{R}^{N \times 3}$ have full column rank. The unique minimizer of $\|A\beta - s\|_2^2$ over $\beta \in \mathbb{R}^3$ satisfies $A^\top A\,\beta = A^\top s$.*

Proof. Define $f(\beta) = \|A\beta - s\|_2^2 = (A\beta - s)^\top (A\beta - s)$. This is a convex quadratic. Setting the gradient to zero gives $\nabla_\beta f = 2A^\top(A\beta - s) = 0$, hence $A^\top A\beta = A^\top s$. Since A has full column rank, $A^\top A$ is positive definite [Strang, 2006], hence invertible, so $\beta^\star = (A^\top A)^{-1}A^\top s$ is the unique solution. The Hessian $\nabla^2 f = 2A^\top A$ is positive definite, confirming $\beta^\star$ is the unique global minimizer [Boyd and Vandenberghe, 2004]. □

Remark 1. *The Kasa method minimizes an algebraic distance, not the geometric (orthogonal) distance from each point to the circle. Under moderate noise with points distributed around the full perimeter, the two are close [Kasa, 1976].*

4.3 Ellipse Fitting by Direct Constrained Least Squares

We use the Fitzgibbon–Pilu–Fisher direct least-squares method [Fitzgibbon et al., 1999]. A general conic section is $ax^2 + bxy + cy^2 + dx + ey + f = 0$ with coefficient vector $\mathbf{a} = (a, b, c, d, e, f)^\top$. The method enforces the ellipticity constraint $4ac - b^2 > 0$ during fitting, guaranteeing the result is always an ellipse.

Partition the $N \times 6$ design matrix as $D = [D_1 \mid D_2]$, where $D_1 \in \mathbb{R}^{N\times 3}$ has columns $[x_i^2,\, x_i y_i,\, y_i^2]$ and $D_2 \in \mathbb{R}^{N\times 3}$ has columns $[x_i,\, y_i,\, 1]$. Form the 3×3 scatter matrices $S_1 = D_1^\top D_1$, $S_2 = D_1^\top D_2$, $S_3 = D_2^\top D_2$. Define $T = -S_3^{-1}S_2^\top \in \mathbb{R}^{3\times 3}$, $M = S_1 + S_2 T \in \mathbb{R}^{3\times 3}$, and the constraint matrix encoding the ellipticity condition $4ac - b^2 > 0$:

$$C = \begin{pmatrix} 0 & 0 & 2 \\ 0 & -1 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

The reduced problem is the standard eigenproblem $C^{-1}M\mathbf{v} = \lambda\mathbf{v}$, where $\mathbf{v} = (v_1, v_2, v_3)^\top$ corresponds to the quadratic coefficients (a, b, c). Among all eigenvectors, the method selects the one satisfying $4v_1v_3 - v_2^2 > 0$ (the ellipticity condition on the quadratic part). The full six-element coefficient vector is $\mathbf{a} = (\mathbf{v}^\top, (T\mathbf{v})^\top)^\top$, recovering (a, b, c, d, e, f).

Proposition 2 (Conic discriminant condition for ellipses)**.** *For a general conic $ax^2 + bxy + cy^2 + dx + ey + f = 0$, if $b^2 - 4ac < 0$ and the curve is nonempty and nondegenerate, then the curve is an ellipse.*

Proof. Let $Q = \left(\begin{smallmatrix} a & b/2 \\ b/2 & c \end{smallmatrix}\right)$ be the matrix of the quadratic form. The condition $b^2 - 4ac < 0$ is equivalent to $\det(Q) = ac - b^2/4 > 0$. Since Q is real symmetric with positive determinant, its eigenvalues λ_1, λ_2 satisfy $\lambda_1\lambda_2 > 0$, so they share the same sign. By the spectral theorem, there exists an orthogonal U such that $U^\top QU = \mathrm{diag}(\lambda_1, \lambda_2)$. Under the change of variables $(x, y)^\top = U(u, v)^\top$, the cross term vanishes: $\lambda_1 u^2 + \lambda_2 v^2 + (\text{linear terms}) + g = 0$. Completing the square gives $\lambda_1(u - u_0)^2 + \lambda_2(v - v_0)^2 = k$. Since λ_1, λ_2 share the same sign and the curve is nonempty, k has the same sign as the eigenvalues, making the level set bounded—an ellipse [Semple and Kneebone, 1952]. □

Conversion to geometric parameters. The function `cart_to_pol` converts conic coefficients (a, b, c, d, e, f) to center (x_0, y_0), semi-major axis a_p, semi-minor axis b_p, and rotation angle ϕ. The center is the stationary point of the quadratic form, found by setting the gradient to zero:

$$\begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = -\begin{pmatrix} d \\ e \end{pmatrix}.$$

The implementation uses halved linear coefficients $b' = b/2$, $d' = d/2$, $e' = e/2$ (with the constant term denoted $g = f$), giving the explicit solution $x_0 = (c\,d' - b'\,e')/(b'^2 - ac)$, $y_0 = (a\,e' - b'\,d')/(b'^2 - ac)$. The semi-axes are extracted from the eigenvalues of $Q = \left(\begin{smallmatrix} a & b' \\ b' & c \end{smallmatrix}\right)$ after centering, and the rotation angle is $\phi = \frac{1}{2}\arctan\left(\frac{2b'}{a-c}\right)$ with appropriate quadrant correction.

4.4 Square Fitting via Empirical Circle-to-Square Conversion

The square fitting procedure exploits a key observation: when the Kasa circle fitter is applied to data that actually comes from a square, it still produces a well-defined radius, and that radius bears a consistent, predictable relationship to the true side length.

Empirical calibration. To discover this relationship, we generated over one thousand known square point clouds with varying side lengths (12 points per side uniformly distributed along the perimeter), applied the Kasa circle fitter to each, and recorded the ratio $\hat{r}/s$ where $\hat{r}$ is the fitted circle radius and s is the true side length. Across all trials, the mean ratio was 0.56985955 with a 95% confidence interval that was stable to the third decimal place. This consistency arises because uniform perimeter sampling of a square creates a reproducible average distance from center, regardless of scale.

The fitting procedure. Given a point cloud P:

1. **Centering.** Center P around the Kasa circle-fit center (x_0^c, y_0^c).
2. **Radius estimation.** On the centered data, sweep candidate radii from the minimum to maximum point distance (step 0.001) and select the radius $\hat{\rho}$ minimizing total absolute radial deviation: $\hat{\rho} = \arg\min_\rho \sum_i \left|\sqrt{x_i^2 + y_i^2} - \rho\right|$.
3. **Side length recovery.** Divide: $\hat{s} = \hat{\rho} \,/\, 0.56985955$.
4. **Rotation search.** Holding $\hat{s}$ fixed, sweep rotation angles from 0° to 360° in steps of 0.1° and select the angle minimizing the piecewise point-to-square residual (Section 4.6).

The constant 0.56985955 is specific to our sampling scheme (12 points per side, uniformly spaced) and would need recalibration for different sampling densities.

4.5 Rectangle Fitting via PCA-Initialized Optimization

Rectangle fitting estimates three parameters simultaneously: width w, height h, and rotation angle ϕ. The point cloud is centered around the ellipse-fit center (x_0^e, y_0^e), since the ellipse fit better captures asymmetry in elongated shapes.

PCA initialization and angle extraction. The initial rectangle orientation is determined by principal component analysis [Hotelling, 1933]. Given centered data $\{p_i^e\}$, we compute the 2×2 sample covariance matrix

$$\Sigma = \frac{1}{N} \sum_{i=1}^{N} p_i^e \, (p_i^e)^\top$$

and find its eigendecomposition. The leading eigenvector $v_1 = (v_{1x}, v_{1y})^\top$ points in the direction of maximum variance. The initial angle is

$$\phi_0 = \arctan 2(v_{1y},\ v_{1x}),$$

which gives the angle of the first principal component measured counterclockwise from the positive x-axis.

Proposition 3 (PCA maximizes projected variance)**.** *Let $\{x_i\}_{i=1}^N \in \mathbb{R}^2$ be centered and let $\Sigma = \frac{1}{N}\sum_i x_i x_i^\top$. The unit vector maximizing $u^\top \Sigma u$ subject to $\|u\|_2 = 1$ is an eigenvector of Σ corresponding to its largest eigenvalue.*

Proof. Since Σ is symmetric positive semidefinite, it has an orthonormal eigenbasis $\{v_1, v_2\}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq 0$. Any unit vector can be written $u = \alpha_1 v_1 + \alpha_2 v_2$ with $\alpha_1^2 + \alpha_2^2 = 1$. Then

$$u^\top \Sigma u = \lambda_1 \alpha_1^2 + \lambda_2 \alpha_2^2 \leq \lambda_1(\alpha_1^2 + \alpha_2^2) = \lambda_1,$$

with equality if and only if $\alpha_2 = 0$, i.e., $u = \pm v_1$ [Hotelling, 1933]. □

Proposition 4 (The PCA angle recovers the direction of maximum spread)**.** *Let $v_1 = (v_{1x}, v_{1y})^\top$ be the unit eigenvector of Σ corresponding to λ_1. Then $\phi_0 = \arctan 2(v_{1y}, v_{1x})$ maximizes the sample variance of the projections of all points onto the line through the origin at angle ϕ_0.*

Proof. The projection of x_i onto $u = (\cos\phi, \sin\phi)^\top$ has value $z_i = u^\top x_i$. The sample variance (with mean zero since data are centered) is

$$\mathrm{Var}(z) = \frac{1}{N}\sum_i (u^\top x_i)^2 = u^\top \left(\frac{1}{N}\sum_i x_i x_i^\top\right) u = u^\top \Sigma\, u.$$

By Proposition 3, this is maximized when $u = v_1$. Since $v_1 = (v_{1x}, v_{1y})^\top$, the maximizing angle is $\phi_0 = \arctan 2(v_{1y}, v_{1x})$. □

For a rectangle, the direction of maximum spread aligns with the longer axis, making this a natural initialization. The initial width and height are estimated as the range of the data projected onto the two PCA axes.

Proposition 5 (Orthogonal projection minimizes distance)**.** *Let $U \subset \mathbb{R}^2$ be a one-dimensional subspace and Π_U the orthogonal projector onto U. For any $x \in \mathbb{R}^2$, $\Pi_U x = \arg\min_{y \in U} \|x - y\|_2$.*

Proof. Write $x = \Pi_U x + r$ where $r \perp U$. For any $y \in U$: $\|x - y\|^2 = \|\Pi_U x - y\|^2 + \|r\|^2 \geq \|r\|^2$, with equality iff $y = \Pi_U x$ [Axler, 2015]. □

Nelder–Mead optimization. Starting from (w_0, h_0, ϕ_0), we minimize the mean squared point-to-rectangle distance using the Nelder–Mead simplex method [Nelder and Mead, 1965] with tolerances `xatol` $= 0.01$, `fatol` $= 0.001$, and `maxiter` $= 300$. Multiple starting angles are tried: if the PCA-estimated aspect ratio is below 1.5, four starts at $\phi_0, \phi_0 + \pi/4, \phi_0 + \pi/2, \phi_0 + 3\pi/4$; otherwise two starts at ϕ_0 and $\phi_0 + \pi/2$. The result with the smallest objective is retained.

4.6 Distance-to-Boundary Residuals

Circle. Radial distances from the Kasa-fit center: $R_\circ = \sum_i \left|\sqrt{(x_i - x_0^c)^2 + (y_i - y_0^c)^2} - \hat{r}\right|$.

Ellipse. Each point is transformed into the ellipse's local frame by translating to center (x_0^e, y_0^e) and rotating by $-\phi_e$. The distance from each transformed point (u_i, v_i) to the axis-aligned ellipse is found by minimizing $(u_i - a_p \cos\theta)^2 + (v_i - b_p \sin\theta)^2$ over θ via scalar numerical minimization.

Square and rectangle. For a shape with axis-aligned half-extents (h_x, h_y) centered at the origin, each point $\tilde{p} = (\tilde{x}, \tilde{y})$ in the shape's rotated frame has distance:

$$d_{\text{rect}}(\tilde{p},\, h_x,\, h_y) = \begin{cases} |\tilde{x} - \text{sign}(\tilde{x})\, h_x| & \text{outside a side, horizontally,} \\ |\tilde{y} - \text{sign}(\tilde{y})\, h_y| & \text{outside a side, vertically,} \\ \sqrt{(|\tilde{x}| - h_x)^2 + (|\tilde{y}| - h_y)^2} & \text{outside a corner,} \\ \min(h_x - |\tilde{x}|,\ h_y - |\tilde{y}|) & \text{inside the shape.} \end{cases}$$

Remark 2 (Distance computation methods)**.** *The residual definition in Section 3 uses the true point-to-boundary distance $d(p, \partial S)$. In practice: the circle residual is the absolute radial deviation (which equals the true distance for circles); the ellipse residual uses numerical minimization over the parametric angle (yielding the true distance); and the rectangle/square residual uses a piecewise signed-distance formula (also exact). All four residuals thus compute the true Euclidean point-to-boundary distance.*

4.7 Derivative-Based Straightness Analysis

Raw residuals alone can fail to distinguish curved from rectilinear shapes under noise. We compute two auxiliary diagnostics.

Point ordering. Points are sorted by polar angle about their centroid, producing an approximate boundary traversal.

Sectional slope analysis. The sorted point cloud is divided into $K = 4$ equal sections. Within each section, finite-difference slopes $m_j = (y_{j+1} - y_j)/(x_{j+1} - x_j)$ are computed between each pair of consecutive points. The key insight is that along a straight edge, the slope between consecutive points stays approximately constant (low variance), while along a curve it changes continuously (high variance). We do not care *what* the slope is—only whether it *changes*.

Pairs with $|\Delta x| < 10^{-6}$ are classified as vertical; those with $|\Delta y| < 10^{-6}$ as horizontal. Each section receives a straightness score between 0 and 1:

- If $\geq 80\%$ of pairs are vertical or $\geq 80\%$ are horizontal: score $= 1.0$.
- If $\geq 80\%$ are diagonal, the coefficient of variation of the slopes is computed: $\text{CV}_k = \sigma_k \,/\, (|\bar{m}_k| + 0.01)$. The score is 1.0 if $\text{CV}_k < 0.5$, 0.0 if $\text{CV}_k > 2.0$, and linearly interpolated otherwise.
- If the section has mixed edge types: score $= 0.0$.

The overall straightness is $\text{ST}(P) = \frac{1}{K}\sum_{k=1}^{K} \text{score}_k \in [0, 1]$.

Proposition 6 (Scale invariance of the CV-based straightness score)**.** *Under uniform scaling $(x, y) \mapsto (sx, sy)$ with $s > 0$, the slope samples m_j are invariant, hence CV_k and $\text{ST}(P)$ are invariant.*

Proof. Under scaling, $\Delta x_j \mapsto s\,\Delta x_j$ and $\Delta y_j \mapsto s\,\Delta y_j$, so $m'_j = s\,\Delta y_j/(s\,\Delta x_j) = m_j$. Since slopes are unchanged, so are their mean, standard deviation, and coefficient of variation [Everitt, 2002]. □

Corner score. Direction angles $\alpha_j = \arctan 2(\Delta y_j, \Delta x_j)$ are computed for consecutive pairs. Direction changes exceeding 60° are counted. The corner score is $\mathrm{CS}(P) = \min(\mathrm{count}/4, 1.0)$.

4.8 Adjusted Residual Scoring

Let $\mathrm{ST}(P)$ be the straightness score and $\mathrm{CT}(P) = 1 - \mathrm{ST}(P)$ the curvature score. The adjusted score is $\widetilde{R}(\mathcal{F}; P) = R(\mathcal{F}; P) - B(\mathcal{F}; P)$, where the bonus B rewards hypothesis–data consistency at weight $w = 0.3$:

$$
\begin{aligned}
B(\circ; P) &= \mathrm{CT}(P) \cdot \hat{r} \cdot w, \\
B(\mathrm{e}; P) &= \mathrm{CT}(P) \cdot \tfrac{\hat{a}_p + \hat{b}_p}{2} \cdot w, \\
B(\square; P) &= \mathrm{ST}(P) \cdot \hat{s} \cdot w, \\
B(\mathrm{r}; P) &= \mathrm{ST}(P) \cdot \tfrac{\hat{w} + \hat{h}}{2} \cdot w.
\end{aligned}
$$

The bonus is proportional to the shape's characteristic size, making it self-normalizing. Since the bonus scales with the fitted shape's dimension, it is possible for $B > R$ in some regimes, yielding negative adjusted scores; only relative comparisons matter, as we take the argmin.

5 Random Forest Baseline

For comparison, we train a Random Forest classifier [Breiman, 2001] on 13 geometric features extracted from the same (x, y) coordinate input: bounding-box aspect ratio, coefficient of variation of centroid distances, distance range ratio, mean/standard deviation/maximum of inter-point direction changes, proportion of direction changes exceeding 45°, circularity ($4\pi A/L^2$), compactness (A/L^2), convex hull fill ratio, section straightness, coefficient of variation of mean radii across 8 angular sectors, and raw corner count. Training uses 2,160 shapes (60 per noise level $\times$ 36 levels, seed 123); testing uses 3,600 shapes (100 per noise level, seed 42). The classifier uses 100 trees, max depth 10, and `StandardScaler` normalization.

6 Experiments

6.1 Synthetic Benchmark

We generate 3,600 shapes across 36 noise levels in $[0.00, 0.70]$ (step 0.02), with 100 samples per noise level (seed 42). Dimensions are drawn uniformly from $[3, 8]$, centers from $[-10, 10]^2$, rotations from $[-180^\circ, 180^\circ]$. Point counts: 50 for circles/ellipses, 48 for squares (12 per side), 50 proportionally distributed for rectangles. Noise is additive uniform, scaled by the shape's characteristic dimension.

6.2 Results

Figure 1 shows the deterministic method's family-matching accuracy from a standalone evaluation run. Accuracy remains at 100% for all noise levels up to and including 0.20 and degrades gradually beyond that. Figure 2 compares both methods on a separate run, where both methods are tested on the exact same 3,600 shapes. A key finding is that **the deterministic method outperforms the Random Forest at all noise levels below approximately 0.32**, achieving higher or equal accuracy throughout the low-to-moderate noise regime. The Random Forest's advantage appears only at higher noise levels, where it benefits from having seen labeled examples during training, and its overall advantage (92.6% vs. 84.8%) is driven primarily by the high-noise regime.

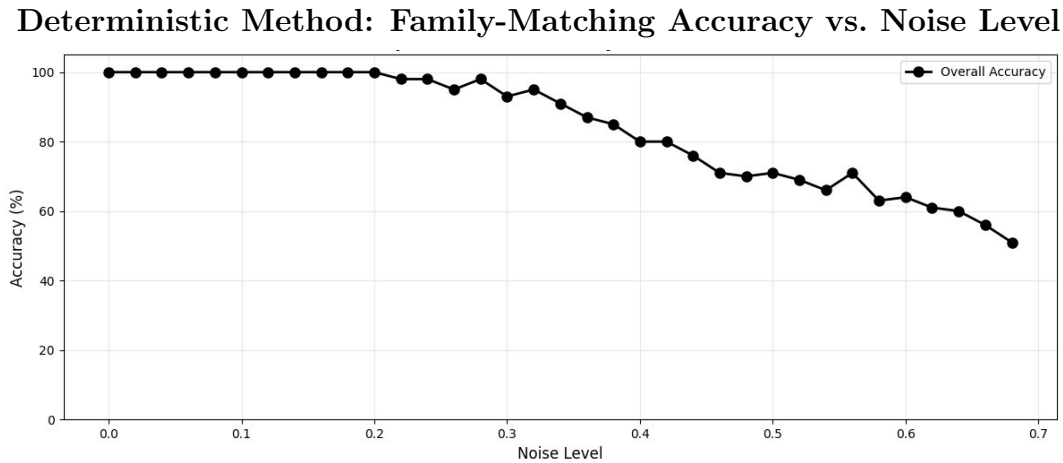

Figure 1: Family-matching accuracy of the deterministic method as a function of noise level, from a standalone run using the same hyperparameters as the comparison experiment but evaluated independently on different random data. Accuracy remains at 100% for noise ≤ 0.20 and degrades gradually thereafter.

Interpreting the noise levels. Since noise is expressed as a fraction of the shape's characteristic dimension, a noise level of 0.30 means each point is displaced by up to 30% of the shape's size. At this level, a square with side length 5 has each boundary point jittered by up to 1.5 units—the resulting point cloud is barely recognizable as a square to a human observer. By noise level 0.50, the boundary signal is almost entirely destroyed. In real-world measurement scenarios—manufacturing tolerances, sensor noise in robotics, measurement error in scientific imaging—noise levels of 0.20 or below are far more typical, placing the deterministic method's perfect-accuracy regime squarely within the practical operating range.

7 Discussion

Strong low-noise performance. The most notable result is that the deterministic method achieves 100% family-matching accuracy up to noise level 0.20 and outperforms the Random Forest at all noise levels below ≈ 0.32. This is remarkable because the Random Forest has access to labeled training data while our method uses none. The geometric fitting approach has a structural advantage at low noise: when the boundary signal is clear, fitting the correct model produces a residual that is unambiguously lower than all competitors, and no amount of training data can improve upon perfect discrimination.

Practical noise regime. The Random Forest's aggregate advantage comes entirely from the high-noise regime ($\eta > 0.32$). But at these noise levels, the shapes are barely distinguishable even visually. A square with 40% noise looks like a formless scatter of points; asking any method to classify it is pushing the limits of the information present in the data. In virtually all practical settings—CNC-machined parts with sub-millimeter tolerances, satellite images with known sensor noise, medical imaging with characterized point-spread functions—the operative noise levels are well within the regime where our method performs perfectly.

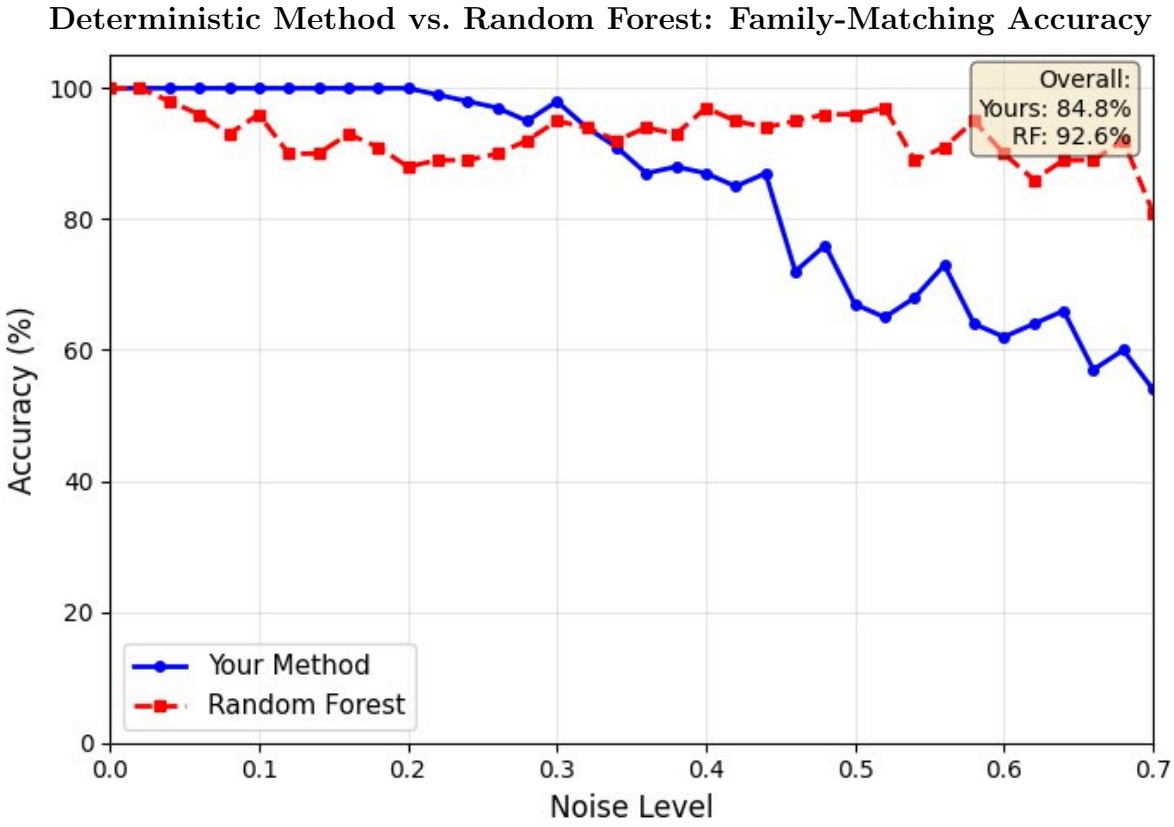

Figure 2: Family-matching accuracy vs. noise for the deterministic method (blue) and the Random Forest (red), evaluated on the *same* set of 3,600 test shapes (seed 42). This is a separate run from Figure 1 using identical hyperparameters but different random data, so the deterministic method's curve here differs slightly from Figure 1. Overall family-matching accuracies are 84.8% and 92.6%, respectively. The deterministic method outperforms the Random Forest at all noise levels below approximately 0.32.

Advantages of the deterministic approach. The method (i) requires no labeled data, (ii) is fully deterministic and reproducible, (iii) outputs rich geometric parameters (center, dimensions, rotation) for every candidate model, (iv) makes all four residuals available for inspection, making failures interpretable, and (v) operates on raw (x, y) coordinate pairs alone.

Limitations. The empirical radius-to-side factor (0.56985955) is tied to our sampling density and would need recalibration for different point distributions. The straightness bonus weight ($w = 0.3$) was not systematically cross-validated. At very high noise, polar-angle sorting can fail, degrading the straightness computation. The Kasa and algebraic ellipse fits minimize algebraic rather than geometric distances.

Future work. The radius-to-side factor could be derived analytically. The bonus weight could be cross-validated or replaced with a model-selection criterion. Robust fitting (Huber loss, RANSAC [Fischler and Bolles, 1981]) could improve high-noise performance. The candidate set could be extended to triangles, regular polygons, or superellipses.

8 Conclusion

We presented a training-free, multi-hypothesis residual method for classifying noisy 2D perimeter point clouds—given only as (x, y) coordinate pairs—among four shape families. The pipeline fits circles via Kasa algebraic least squares, ellipses via Fitzgibbon–Pilu–Fisher constrained least squares, squares via an empirically calibrated circle-to-square conversion with rotation search, and rectangles via PCA-initialized Nelder–Mead optimization. A scale-invariant, derivative-based straightness score rewards agreement between each candidate's boundary type and the data. On 3,600 synthetic shapes across 36 noise levels, the method achieves 100% accuracy up to noise level 0.20, outperforms a Random Forest at all noise levels below 0.32, and achieves 84.8% family-matching accuracy overall versus 92.6% for the Random Forest. Since noise levels above 0.30 produce shapes that are largely indistinguishable even to a human observer, the deterministic method provides strong performance across the entire practical noise regime while requiring no training data and producing interpretable geometric parameters.

Reproducibility

Dataset. 3,600 shapes: 100 per noise level, 36 levels in $[0.00, 0.70]$ (step 0.02). Test seed 42; training seed 123. Dimensions in $[3, 8]$, centers in $[-10, 10]^2$, rotations in $[-180°, 180°]$. Point counts: 50 (circles/ellipses), 48 (squares), 50 proportionally distributed (rectangles). Noise is additive uniform, scaled by the shape's characteristic dimension.

Hyperparameters. Radius sweep step: 0.001. Rotation sweep step: 0.1°. Straightness bonus weight: $w = 0.3$. Stabilization: $\varepsilon = 10^{-6}$ (slope), $\delta = 0.01$ (CV denominator). Nelder–Mead: `xatol` $= 0.01$, `fatol` $= 0.001$, `maxiter` $= 300$. Multi-start threshold: PCA aspect ratio $< 1.5 \Rightarrow 4$ starts; else 2. Random Forest: 100 trees, max depth 10, `StandardScaler`. Empirical radius-to-side factor: 0.56985955.

Libraries. `numpy.linalg.solve` (circle); `numpy.linalg.eig` (ellipse); `scipy.optimize.minimize` (ellipse distance, rectangle); `sklearn.decomposition.PCA` (rectangle); `sklearn.ensemble.RandomForestClassif` (baseline).

References

Sheldon Axler. *Linear Algebra Done Right.* Springer, 3rd edition, 2015. doi: 10.1007/978-3-319-11080-6.

Stephen Boyd and Lieven Vandenberghe. *Convex Optimization.* Cambridge University Press, 2004.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.

Brian S. Everitt. *The Cambridge Dictionary of Statistics.* Cambridge University Press, 2nd edition, 2002.

Martin A. Fischler and Robert C. Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981. doi: 10.1145/358669.358692.

Andrew W. Fitzgibbon, Maurizio Pilu, and Robert B. Fisher. Direct least squares fitting of ellipses. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 21(5):476–480, 1999. doi: 10.1109/34.765658.

Harold Hotelling. Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology*, 24(6):417–441, 1933. doi: 10.1037/h0071325.

Ming-Kuei Hu. Visual pattern recognition by moment invariants. *IRE Transactions on Information Theory*, 8(2):179–187, 1962. doi: 10.1109/TIT.1962.1057692.

I. Kasa. A circle fitting procedure and its error analysis. *IEEE Transactions on Instrumentation and Measurement*, IM-25(1):8–14, 1976. doi: 10.1109/TIM.1976.6312298.

J. A. Nelder and R. Mead. A simplex method for function minimization. *The Computer Journal*, 7(4):308–313, 1965. doi: 10.1093/comjnl/7.4.308.

Joseph O'Rourke. *Computational Geometry in C*. Cambridge University Press, 2nd edition, 1998.

Vaughan Pratt. Direct least-squares fitting of algebraic surfaces. *ACM SIGGRAPH Computer Graphics*, 21(4):145–152, 1987. doi: 10.1145/37402.37420.

Charles R. Qi, Hao Su, Kaichun Mo, and Leonidas J. Guibas. PointNet: Deep learning on point sets for 3D classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 652–660, 2017.

J. G. Semple and G. T. Kneebone. *Algebraic Projective Geometry*. Oxford University Press, 1952.

Gilbert Strang. *Linear Algebra and Its Applications*. Thomson Brooks/Cole, 4th edition, 2006.

Gabriel Taubin. Estimation of planar curves, surfaces, and nonplanar space curves defined by implicit equations with applications to edge and range image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 13(11):1115–1138, 1991. doi: 10.1109/34.103273.